STAT2004 – Tutorial Week 13

Exercise 1 (Comparing four control groups in a larger study[1])

A large study investigating the use of carbon tetrachloride as a worm killer is described in Armitage (1983). In this question, we focus only on a subset of the experiment involving four control groups. If there are differences in these four control groups, then this may be attributable to variations in the experimental conditions, lab settings, lab technicians, etc.. and so a more carefully controlled experiment with clearer protocols may be needed for follow-up studies.

The number of worms in each of five rats under each of four control groups at the end of the study are tabulated below:

Group I	Group II	Group III	Group IV
279	378	172	381
338	275	335	346
334	412	335	340
198	265	282	471
303	286	250	318

The dataset can be loaded in `R` using the following commands:

```
# input the data as a single vector
worms = c(279, 378, 172, 381, 338, 275, 335, 346, 334, 412, 335, 340,
          198, 265, 282, 471, 303, 286, 250, 318)

# group labels should follow the order of the observations:
group = rep(c("I","II","III","IV"), 5)
```

A plot of the data can be generated via the command:

```
# use the "lattice" library for some cleaner graphics
library(lattice)
dotplot(worms ~ group, xlab = "Group", ylab = "No. of worms",
        main = "Number of worms in rats in each of four control groups")
```

(a) Argue why the side-by-side dotplot generated above is more appropriate for these data than a side-by-side boxplot.

(b) Visually assess the validity of the "common variance" assumption that underpins ANOVA.

(c) Test whether there are any mean difference between the four control groups. Write down an underlying model for the data, clearly state the null and alternative hypotheses in terms of the model parameters, compute a test statistic and make your decision based on a p-value.

A rival statistician notes that the sample mean of Group IV (which was $\overline{y}_{4\bullet} = 371.2$) appears much larger than the sample mean of Group III (which was $\overline{y}_{3\bullet} = 274.8$). The two sample standard deviations were $s_4 = 60.20$ and $s_3 = 67.97$, respectively. This inspires the rival statistician to carry out a *two-sample t-test* on just these two groups of rats.

(d) Replicate the rival statistician's approach by carrying out a two-sample t-test on only groups IV and III. What is the p-value of this test, and what would be the rival statistician's conclusion?
(e) How do we reconcile the apparent contradiction between our results from part (c) and part (d)? Suggest a quick-and-dirty way to adjust one of these approaches appropriately.

[1] Adapted from Problem 12.19 of *Mathematical Statistics & Data Analysis* (Rice, 1995)

Exercise 2 (Laptops and scrotal temperatures)

Studies have suggested that heating the scrotum by just 1 degree Celsius can temporarily reduce sperm count and sperm quality, while a 2 degree increase can cause long-term irreversible damage.

A recent study by Sheynkin et al. (2011) examined the effect of using a laptop computer on the lap on the changes in scrotal temperatures. To emulate this experiment, a total of 150 male participants were randomly assigned to one of the three sitting conditions (legs together, legs apart, and legs together with a laptop cooling pad). Each participant was asked to use the same laptop to watch the same video for 60 minutes, after which the changes in scrotal temperatures were recorded. The research question is to test if there any differences in scrotal temperatures between the three sitting conditions.

The full dataset can be downloaded from Blackboard > Learning Resources > Tutorials > scrotaltemp.csv. You can read the dataset into `R` by setting the working directory to where you saved the file and using the following command:

```
scrotaltemp = read.csv("scrotaltemp.csv", header=T)
```

(a) Produce an appropriate plot of the data. Briefly explain why your plot provides a good visual summary of the data.
(b) Address the research question by carrying out an appropriate test. Write down an underlying model for the data, clearly state the null and alternative hypotheses in terms of the model parameters, compute a test statistic and make a decision based on a p-value.
(c) Critically assess any assumptions you made in your analysis from part (b).
(d) In light of your findings from parts (a)–(c), and any other analyses you deem relevant, which of the three sitting conditions would you recommend to a male friend who is going to be using a laptop computer on his lap to watch a movie tonight?

Exercise 3 (Two-way ANOVA of survival times by poison and treatments[2])

The following table gives the survival times (in hours) for animals in an experiment whose design consisted of three poisons, four treatments, and four replications per experimental setting:

	Treatment							
Poison	A		B		C		D	
I	3.1	4.5	8.2	11.0	4.3	4.5	4.5	7.1
	4.6	4.3	8.8	7.2	6.3	7.6	6.6	6.2
II	3.6	2.9	9.2	6.1	4.4	3.5	5.6	10.0
	4.0	2.3	4.9	12.4	3.1	4.0	7.1	3.8
III	2.2	2.1	3.0	3.7	2.3	2.5	3.0	3.6
	1.8	2.3	3.8	2.9	2.4	2.2	3.1	3.3

The dataset can be loaded in `R` using the following commands:

```
# input all survival times as a single vector
survivaltime = c(3.1,4.5,4.6,4.3,8.2,11.0,8.8,7.2,4.3,4.5,6.3,7.6,4.5,7.1,6.6,6.2,
                 3.6,2.9,4.0,2.3,9.2,6.1,4.9,12.4,3.1,4.0,2.3,2.5,7.1,3.8,3.0,3.6,
                 2.2,2.1,1.8,2.3,3.0,3.7,3.8,2.9,2.3,2.5,2.4,2.2,3.0,3.6,3.1,3.3)

# define treatment levels in same order as observations
Treatment = rep(c(rep("A",4),rep("B",4),rep("C",4),rep("D",4)),3)
# define poison levels in same order as observations
Poison = c(rep("I",16),rep("II",16),rep("III",16))
```

A plot of the data can be generated via the command:

```
# use "lattice" library for cleaner graphics
library(lattice)
dotplot(survivaltime ~ Treatment | Poison, layout = c(3,1),
        ylab="Survial time (hours)", xlab="Treatment")
```

(a) Argue why the side-by-side(-by-side) dotplot generated above is more appropriate for these data than a side-by-side(-by-side) boxplot.

(b) Carry out a two-way ANOVA on these data to test the effects of the two main factors and their interaction.

(c) Critically assess the assumptions behind ANOVA in the context of these data.

(d) Box and Cox (1964) suggested that the reciprocal of the survival time can be interpreted as a rate of death. Conduct a two-way ANOVA on the reciprocals, and compare to the results from part (b).

(e) Critically assess the assumptions behind ANOVA when using the reciprocals.

(f) Which of the two analyses is more appropriate for these data? Briefly justify your choice.

[2] Adapted from Problem 12.27 of *Mathematical Statistics & Data Analysis* (Rice, 1995)

Exercise 4 (Two-way ANOVA with interactions as a linear model)

Recall that a linear model is any model than can be written in the form

$$Y_i = X_i^\top \boldsymbol{\beta} + \epsilon_i \ ,$$

where X_i is a vector of covariates for each observation i, $\boldsymbol{\beta}$ is a vector of regression parameters, and each ϵ_i is a some random variable with mean 0.

We can also write this in vector-matrix form via

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where $\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_n)^\top$ is a single vector of all the responses, $\boldsymbol{X}$ is a matrix of covariates (called the *design matrix*), and $\boldsymbol{\epsilon}$ is a vector of all the random error terms.

(a) Write the two-way ANOVA model with interactions from Exercise 3 (b) in the form of a linear model. Clearly describe what the covariate matrix $\boldsymbol{X}$ looks like, identify each component of the parameter $\boldsymbol{\beta}$, and state the distribution of the error vector $\boldsymbol{\epsilon}$.